EB-H-JEPA: Energy-Based Hamiltonian JEPA World Models

A Failure Atlas and a Controlled Sample-Efficiency Test in Crafter

Sehaj Randhir Singh
NYU Tandon School of Engineering
sehaj.randhir.singh@nyu.edu

Abstract

World models that operate in a learned latent space (JEPA; LeCun 2022) are the leading bet for sample-efficient, long-horizon model-based RL (DreamerV3, Hafner et al. 2023). A recurring hypothesis is that imposing physical structure on the latent predictor — Hamiltonian, metriplectic, or GENERIC — should make learned dynamics faithful and stable. We test this hypothesis twice, with instruments of increasing fidelity. First, on a nonlinearly lifted Lorenz-63 system, we audit five structurally distinct latent predictors under a chaos-aware protocol. The result is a failure atlas: each constraint class fails, but on a property disjoint from the one it enforces. A rigid Hamiltonian predictor freezes and cannot dissipate; a naive metriplectic predictor sits on a dead $LL^\top$ saddle with dissipation pinned at exactly zero; fixing both bugs (constant divergence-free J, nonzero R initialization) makes dissipation engage and recovers the true phase-space contraction -13.667 to $\sim 1\%$ accuracy — yet spectral alignment still cannot pin individual Lyapunov exponents. Second, and constructively, we insert the fixed-metriplectic predictor into a compact DreamerV3-style agent and measure sample efficiency on Crafter (Hafner 2021), the benchmark of the world-model group that introduced RSSM. If structure helps, this is the first positive evidence that physical inductive biases pay off in an RL world model; if it hurts, that is the honest negative the field needs. All code, runs, and evidence are public; every claim traces to a JSON artifact.

1 Introduction

JEPA-style world models [12, 2, 5] and their model-based RL instantiations [9, 10, 3] share a core bet: a latent predictor learned from observations can model dynamics that stay stable and meaningful over long horizons. The predictor is the crux. DreamerV3's categorical RSSM learns it from data alone; a growing line of work asks whether injecting physical structure — symplectic/Hamiltonian form [6, 4, 16], metriplectic/GENERIC dissipation [15, 17, 11], or learned divergence gates [7] — yields predictors that are more faithful and more sample-efficient.

This paper reports a two-part test of that hypothesis, built to be reproducible end to end.

Part 1: a failure atlas on a known chaotic system. We train five structurally distinct latent predictors on Lorenz-63 lifted into a 64-D latent chart and audit each under a chaos-aware protocol: rollout boundedness, leading Lyapunov exponent, phase-space contraction, attractor overlap, and dissipation engagement. The honest headline is negative, and more precise than "structure doesn't help": *every constraint class fails, but on a property disjoint from the one it enforces.* A rigid Hamiltonian predictor (fixed canonical J, learned H) is conservative by construction and cannot dissipate; it freezes into a quasi-periodic tube with near-zero motion. A naive metriplectic predictor learns R through $R = LL^\top$ but sits on the dead saddle where $L = 0$: R stays exactly 0, dissipation never engages. Fixing both bugs — constant divergence-free J (closing the learned-skew loophole) and nonzero L initialization (breaking the dead saddle) — makes R engage and recovers the true contraction -13.667 to $\sim 1\%$. Yet even the best arm cannot pin individual Lyapunov exponents: differentiable QR spectral shaping controls only a finite-horizon statistic, not the asymptotic Oseledets spectrum [14]. And overlap metrics reward dead models: a frozen predictor sits inside the encoded attractor and posts the best Chamfer score while being dynamically inert.

Part 2: a controlled sample-efficiency test in Crafter. A failure atlas on Lorenz tells us where structure fails in isolation; it does not tell us whether structure helps or hurts an *actual RL world model*, where the predictor co-adapts with an actor trained by latent imagination. We build a compact but faithful reimplementation of the DreamerV3 recipe — CNN encoder/decoder, recurrent memory, categorical posterior, KL free bits, symlog targets, TD(λ) imagination — in which the stochastic prior is pluggable. The baseline is DreamerV3's exact categorical RSSM; the treatment is a continuous latent whose prior mean advances under the fixed-metriplectic map from Part 1. Everything else is identical. We measure sample efficiency on Crafter [8], the open-world benchmark introduced by the RSSM group itself. The result — whether structure helps, hurts, or is neutral — is a controlled answer to the field's open question, and either way it is publishable. Preliminary CPU verification (Sec. 4.2) confirms both arms train and the pipeline is sound; the headline numbers (Table 2) come from the 1-hour T4 runs whose exact commands are shipped in the repo.

2 Background

JEPA. Joint-Embedding Predictive Architectures [12] predict in representation space, not pixel space: an encoder produces a latent z_t from observation x_t, and a predictor advances it. The EB-JEPA line [2, 5] shows strong representation learning without reconstruction.

DreamerV3. Hafner et al. [9] train a recurrent state-space model (RSSM) — categorical stochastic prior $p(z_t \mid h_t)$, posterior $q(z_t \mid h_t, x_t)$, GRU memory h — jointly with reconstruction, reward, and continue heads, then train an actor-critic by "imagining" rollouts in the learned latent. It is the sample-efficiency reference for visual control [8].

Metriplectic / GENERIC. Dissipative physical systems are not Hamiltonian; they follow the metriplectic (GENERIC) form $\dot{z} = J\nabla H - R\nabla S$ [15], with J antisymmetric (energy-conserving flow), R symmetric positive semidefinite (dissipation), and H, S the energy and entropy potentials. Learned versions [17, 11, 7] parameterize J and R; the two failure modes isolated in this paper — learned J exploiting the skew loophole, and $R = LL^\top$ sitting at the dead $L = 0$ saddle — are exactly the traps a naive parameterization falls into.

3 Method

3.1 Latent predictors (Part 1)

All arms share: a 64-D latent chart (nonlinear diffeomorphic lift of Lorenz-63), an MLP dynamics predictor, SIGReg anti-collapse regularization, and identical training budgets. They differ only in the algebraic class of the predictor:

B Unconstrained MLP $z' = f_\theta(z)$.

C Rigid Hamiltonian $z' = z + dt\, J\nabla_z H(z, h)$, fixed canonical J; conservative, cannot dissipate.

D Naive metriplectic, learned J, $R = L(z)L(z)^\top$, L initialized at zero.

E *Fixed* metriplectic: constant divergence-free J ($\mathrm{div}(J\nabla H) = 0$ exactly), L initialized at 0.05, learned H, S conditioned on GRU state h.

F E plus differentiable-QR spectral alignment on the rollout Jacobian.

Chaos-aware evaluation: rollout boundedness, motion ratio $m = \mathbb{E}|\Delta z|/\mathbb{E}|z|$, leading exponent λ_1 by the QR method, contraction $\mathrm{tr}(R)/n$ vs. true divergence -13.667, and Chamfer/Hausdorff overlap against the encoded attractor.

3.2 Pluggable predictor in a DreamerV3-style agent (Part 2)

The treatment predictor is the Part-1 E-arm embedded in the RSSM:

$$z' = z + dt\big(\underbrace{J\,\nabla_z H(z,h)}_{\text{flow}} - \underbrace{R(z,h)\,\nabla_z S(z,h)}_{\text{dissipation}}\big), \qquad R = LL^\top \succeq 0, \tag{1}$$

with constant canonical J (dimension 64), L initialized at 0.05, dt learnable, and H, S MLPs conditioned on the GRU memory h. The prior is Gaussian with this mean and fixed variance, keeping the KL analytic with free bits. The baseline is DreamerV3's categorical RSSM (32 classes $\times$ 32 codes). All other components — encoder, decoder, reward/continue heads, TD(λ) imagination, actor-critic — are byte-identical across arms; any difference is attributable to the predictor structure alone.

4 Experiments

4.1 Part 1: failure atlas (3 seeds, real numbers)

Table 1 summarizes 3-seed runs (full traces in `results/`, runner in `benchmarks/`).

arm	motion ratio	model λ_1	contraction	$\mathrm{tr}(R)/n$
B unconstrained	—	explodes	—	—
C rigid Hamiltonian	~ 0	~ 0	≈ 0	—
D naive metriplectic	low	overshoot	wrong	0.00000
E fixed metriplectic	1.1–2.9	1.0–4.3	$\mathbf{-13.80 \pm 0.17}$	2.3–6.2
F + spectral align	1.2–3.8	1.4–5.0	-13.25 ± 0.19	2.5–8.3

Table 1: Failure atlas on lifted Lorenz-63. True divergence -13.667; true $\lambda_1 \approx 0.9$. E recovers contraction to $\sim 1\%$, but no arm pins λ_1; the rigid arm posts the best *overlap* while being dynamically dead.

arm	return @ 10k steps	return @ 20k steps
RSSM (DreamerV3)	–	–
Metriplectic (E-arm)	–	–

Table 2: Crafter sample efficiency, to be filled from the T4 runs (see `kaggle/README.md` for the one-click protocol). Placeholders are deliberate: we do not ship numbers we have not run.

Reading. (i) The dissipation gate works: once J is constant and R escapes the dead saddle, learned contraction matches truth. (ii) Spectral shaping does not: pinning the *sum* of the spectrum cannot pin any individual Oseledets exponent. (iii) Overlap metrics reward dead models; they must be gated on rollout liveness.

4.2 Part 2: Crafter sample efficiency (controlled)

Protocol. Two arms (RSSM baseline, metriplectic treatment), identical seeds and budgets; 20k environment steps; world model trained on batches of 16×16-frame sequences; evaluation every 5k steps by greedy rollout return. Runs execute with a hard 55-minute wall-clock budget on a T4 (exact commands: `experiments/crafter/train.py`, `kaggle/README.md`).

Hypotheses. H_+: structure helps — the metriplectic arm needs fewer environment steps to reach a given return, because its prior mean is a better dynamics prior (less posterior collapse, longer useful imagination). H_-: structure hurts — the continuous latent and learned potentials slow the world model relative to the discrete categorical prior. Either outcome is a publishable, controlled answer.

Preliminary verification (CPU, fake env). Both arms train end-to-end from a clean checkout: reconstruction loss falls (metriplectic 0.19, RSSM 0.08 after $\sim$100 steps at reduced batch), KL free bits active, actor/critic losses finite, evaluation episodes complete, JSON metrics written. This validates the pipeline; it is not evidence about the hypothesis. The headline comparison (Table 2) is produced by the shipped T4 runs.

5 Related work

Structure-inducing predictors: Hamiltonian neural networks [6, 4, 16], metriplectic/GENERIC learning [17, 11, 7], latent-world-model physics [1, 13]. JEPA and model-based RL: [12, 2, 5, 9, 10, 3]. This paper differs in two ways: it documents *which* structural failure mode each class exhibits (the atlas), and it runs the first controlled predictor-ablation inside a DreamerV3-style agent on the benchmark (Crafter) of the group that introduced RSSM.

6 Conclusion

Physical structure in latent predictors is not a free lunch: each naive class fails on a specific, diagnosable property. The fixes that make dissipation genuinely engage (constant divergence-free J; broken dead saddle) are concrete and validated. Whether those fixes pay off inside a real RL world model is the open question we answer with the Crafter experiment — shipped, reproducible, and bounded by an hour of compute. The code, runs, and drafts are public; we invite scrutiny of every number.

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